The Open Source Audit

A Capstone Project for OSS NGMC Course

Course: Open Source Software | Unit Coverage: 1 – 5 Max Marks: 100

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1. Introduction

Ive created a project on the topic git for the course open source software (Vityarthi) . In todays time most of the electronic items use open source software, open source software make our work easy and help to perform complex problems or certain tasks

One of the most important open-source tools used today is Git, a system used to manage software development projects. Git allows us to track all the changes made in the code , also helps us to collaborate with other users or developers , and maintain a complete history of their work

The report will include an audit of the open-source project Git. The audit will include information about the origin and philosophy of the project, the project's licensing scheme, its role in the Linux ecosystem, and a comparison of its uses . Five Linux shell scripts will also be created to illustrate the use of open-source tools.

2. Choosing My Software

Approved software options

Software	Category	License	Why it's interesting
Linux Kernel	Operating System	GPL v2	*The foundation everything else runs on*
Apache HTTP Server	Web Server	Apache 2.0	*Powers ~30% of all websites globally*
MySQL	Database	GPL v2 / Commercial	*A license with a dual-model story to tell*
Firefox	Web Browser	MPL 2.0	*A nonprofit fighting for an open web*
VLC Media Player	Multimedia	LGPL / GPL	*Plays anything — built by students in Paris*
LibreOffice	Office Suite	MPL 2.0	*Born from a community fork — a real lesson*
Python	Programming Language	PSF License	*A language shaped entirely by community*
Git	Version Control	GPL v2	*The tool Linus built when proprietary failed him*

My chosen software is “Git”

3. Project Report Structure

Part A — Origin and Philosophy (Units 1 & 2)

A1. The problem Git was created to solve :

Before Git existed, software development teams relied on centralized version control systems such as CVS and Subversion. These systems stored code in a central repository and required developers to constantly connect to a server to update their work. This approach often created problems when teams were large or distributed across different locations.

The creation of Git was closely tied to the development of the Linux kernel. For many years, Linux developers used a proprietary version control system called BitKeeper. However, in 2005 the relationship between the Linux community and the BitKeeper developers broke down, and the Linux project suddenly lost access to the tool.

This created a major problem because thousands of developers around the world were contributing to the Linux kernel. Without a proper version control system, coordinating changes would become extremely difficult.

To solve this problem, Linus Torvalds, the creator of Linux, decided to build a new version control system from scratch. His goal was to create a tool that was fast, reliable, and capable of handling extremely large projects.

Git was designed with several key ideas:

- **Speed and efficiency**
- **Strong data integrity**
- **Distributed collaboration**
- **Ability to manage large projects**

Unlike older systems, Git allows every developer to have a full copy of the repository on their own machine. This means developers can work offline, experiment freely, and merge changes later.

Today Git has become the most widely used version control system in the world and forms the foundation of platforms such as GitHub and GitLab.

A2. The license — what it actually says :

Git is distributed under the GNU General Public License version 2 (GPLv2). This license is one of the most well-known open-source licenses and is designed to ensure that software remains free and open.

The GPL license supports the four freedoms of free software:

1. **The freedom to run the program for any purpose.**
2. **The freedom to study how the program works and modify it.**
3. **The freedom to redistribute copies.**
4. **The freedom to distribute modified versions.**

Git grants all four of these freedoms to users.

However, the GPL license also has an important condition called copyleft. This means that if someone modifies Git and distributes their version, they must also release the modified source code under the same license. This prevents companies from taking open-source code and turning it into proprietary software.

A company can still sell software that includes Git, but they must provide the source code and allow users to modify it.

GPL vs MIT License

The GPL license is stricter than the MIT license. The MIT license allows developers to reuse the code in almost any way, even in proprietary software, without releasing modifications publicly.

If I were creating my own software project, I might choose the MIT license if I wanted maximum flexibility and wide adoption. However, if I wanted to ensure that improvements always remain open-source, I would choose the GPL license.

Free as in Freedom vs Free as in Free Beer

Open-source software is often described using the phrase "free as in freedom, not free as in free beer."

This means the focus is not simply on price, but on the freedom to modify and share the software.

Git is a good example of this concept. It is free to download, but more importantly, developers can modify its source code, contribute improvements, and build tools around it.

A3. The ethics of open source

But open source is more than just a license; it is a philosophy.

It has been argued that all software should be open source because of the inherent advantages in terms of security and innovation. This is because when source code is open source, anyone can view it and correct errors.

But there are some arguments against all software being open source. Companies spend a lot of money on research and development, and by having proprietary software, they can protect their ideas and make money.

Another ethical issue is how some companies make money by using open source software and then selling their own versions of these programs. There are a number of large technology companies that make extensive use of open source software.

The term "standing on the shoulders of giants" refers to how modern technology is a result of past achievements in technology. Open source is a way of achieving this.

Overall, open source is a way of sharing and working together.

Part B — Linux Footprint (Unit 2)

It is tightly integrated with Linux systems, and the installation process is often as easy as installing with the system's package manager.

Installation Method :

In many Linux distributions, the installation process is as follows:

-sudo apt install git

or

-sudo yum install git

Key Directories :

Once installed, the location of the various Git files is as follows:

/usr/bin/git – the main program

/etc/gitconfig – the systemwide config

/usr/share/git – various support files

In the repository, the version history is contained within the repository's hidden directory named .git.

User and Group :

In Linux, the user who runs the program is the user associated with the Git process. In server environments, the process is often run as the git user.

Using separate users enhances the system's security.

Service Management :

Unlike some other programs, Git is not run as a background process. It is run as an interactive process with various commands:

git clone

git commit

git push

git pull

It is possible that the Git hosting sites run the various processes in the background.

Update Model :

Security updates for the program are provided through the Linux system's package manager. The open-source community constantly monitors the system for security breaches, and the fixes are provided as updates through the Linux system's repository.

- ## Part C — The FOSS Ecosystem (Units 3 & 4)

However, no open-source project is an island, and Git is no different.

Dependencies :

Git uses several open-source projects and tools such as:

zlib(a compression library)

OpenSSL (a cryptographic library)

Curl (a data transfer library)

These tools are essential for the secure transfer and storage of data by Git.

Tools Built on Git

Modern development platforms are built on top of Git. Some of these platforms are:

GitHub

GitLab

Bitbucket

Role in Web Development :

Git is an essential tool for modern web development. Git is used for managing code for websites, mobile apps, and server-side infrastructure.

In most development environments, Git is part of the DevOps process. This process allows for the automation of testing and deployment.

Community :

Git has a large global community of developers.

The developers use mailing lists and forums for communication. Companies such as Google, Microsoft, and Facebook are involved in the development of Git. However, the project is open for anyone to contribute to.

- **Part D — Open Source vs Proprietary (Critical Analysis)**

Dimension	Git	Proprietary Alternative
Cost	Free to use	Usually paid
Security	Open to review	Hidden
Support and reliability	Community available	Vendor support
Freedom to modify	Fully customisable	Limited customisation
Community vs corporate	Community driven	Controlled by a company

Final verdict :

Git offers many advantages because of its open nature and distributed structure. It allows programmers to work in collaboration while retaining full control over their code.

Although other version control systems offer support and other integrated features, Git offers more reliability and support, which makes it the most used version control system.

Therefore, Git has emerged as the global standard for version control.

if given an opportunity, I would definitely like to contribute to open-source projects like Git, which can help the entire programming community.

4. Shell Script Tasks

As part of this project, five Linux shell scripts have been developed to illustrate the application of Linux command-line knowledge. The scripts are designed to accomplish a

particular task regarding the Linux system and utilize different concepts of Linux shell scripting programming.

The scripts have been developed and executed on a Linux system to illustrate the application of automation for open-source tools.

Script 1 — System Identity Report

Purpose :

This script will show vital information regarding the Linux system. It is similar to a welcome screen, which will introduce the system environment to the user.

This script will display system information, such as kernel version, current logged-in user, system uptime, and current date, along with a message regarding the open-source license of the Linux operating system.

Concepts Used :

Variables

Command substitution $()

echo command

Basic output formatting

```
==================================
 Open Source Audit — Akshat Bhave
==================================
Chosen Software : Git
Kernel Version  : 23.6.0
User            : akshat
Uptime          :
Date            : Tue Mar 31 01:39:25 IST 2026
Operating System: Darwin
License         : Linux is distributed under the GNU GPL license.
==================================
akshat@Akshats-MacBook-Air-2 oss-audit-24BAI10377 %
```

Script 2 — FOSS Package Inspector

Purpose:

This script is designed to inspect if a particular open-source software package is installed on a system. In this project, we will be using this script to inspect if Git is installed on a system.

If a particular package is installed, we will be able to retrieve information such as the version, license, and a summary of the software.

A case statement will be utilized to print a brief description of the purpose of the software.

Concepts Used :

if-then-else condition

case statement

grep

Package query commands

```
$ script2_package_inspector.sh
#!/bin/bash
# Script 2: FOSS Package Inspector

PACKAGE="git"

if command -v $PACKAGE &> /dev/null
then
    echo "$PACKAGE is installed."
    $PACKAGE --version
else
    echo "$PACKAGE is NOT installed."
fi

case $PACKAGE in
    git) echo "Git: A distributed version control system created by Linus Torvalds." ;;
    apache) echo "Apache: A powerful open-source web server." ;;
    mysql) echo "MySQL: A widely used open-source database." ;;
    vlc) echo "VLC: A free and open-source multimedia player." ;;
    *) echo "Unknown package." ;;
esac
```

Script 2 — FOSS Package Inspector

Purpose

This script is intended to inspect if a particular open-source software package is installed on the system. In this project, we will be using this script to inspect if Git is installed on the system.

If the particular package is installed, it is possible to obtain the version, license, and summary of the particular package.

case statement will be used to print the brief description of the purpose of the particular package.

Concepts Used :

if-then-else condition

case statement

grep

package query commands

```
#!/bin/bash
# Script 3: Disk and Permission Auditor

DIRS=("/etc" "/var/log" "/home" "/usr/bin" "/tmp")

echo "Directory Audit Report"
echo "----------------------"

for DIR in "${DIRS[@]}"
do
    if [ -d "$DIR" ]; then
        PERMS=$(ls -ld $DIR | awk '{print $1, $3, $4}')
        SIZE=$(du -sh $DIR 2>/dev/null | cut -f1)
        echo "$DIR => Permissions: $PERMS | Size: $SIZE"
    else
        echo "$DIR does not exist"
    fi
done
```

Script 4 — Log File Analyzer

Purpose

This script analyzes a given log file and determines the number of times a keyword appears in the file. The analysis of a log file is important in determining system errors.

The script works by reading the file line by line and determining whether the keyword appears in the line.

Concepts Used :

While read loop

If conditions

Counter variables

Command line arguments $1

$ script4_log_analyzer.sh

```bash
#!/bin/bash
# Script 4: Log File Analyzer

LOGFILE=$1
KEYWORD=${2:-error}
COUNT=0

if [ ! -f "$LOGFILE" ]; then
    echo "Error: File not found."
    exit 1
fi

while IFS= read -r LINE
do
    if echo "$LINE" | grep -iq "$KEYWORD"; then
        COUNT=$((COUNT+1))
    fi
done < "$LOGFILE"

echo "Keyword '$KEYWORD' found $COUNT times in $LOGFILE"
```

Script 5 — Open Source Manifesto Generator

Purpose

This script will prompt the user to answer three questions and will then generate a short open source philosophy manifesto based on their answers.

This script will show how to use user input to dynamically generate text.

Concepts Used :

read command for user input

String concatenation

Writing output to a file

date command

```
echo "Answer three questions to generate your manifesto."
echo ""

read -p "1. Name one open-source tool you use every day: " TOOL
read -p "2. In one word, what does 'freedom' mean to you? " FREEDOM
read -p "3. Name one thing you would build and share freely: " BUILD

DATE=$(date '+%d %B %Y')
OUTPUT="manifesto_$(whoami).txt"

echo "On $DATE, I reflected on open source." > $OUTPUT
echo "Tools like $TOOL remind me that software freedom means $FREEDOM." >> $OUTPUT
echo "One day I hope to build $BUILD and share it openly with the community." >> $OUTPUT

echo ""
echo "Manifesto saved to $OUTPUT"
cat $OUTPUT
```

Conclusion :

Open-source software has revolutionized the development and usage of technology worldwide. From this audit of Git, it is evident that open-source software thrives on collaboration, openness, and contributions from a wide variety of individuals. Git was originally designed to help in the development of the Linux kernel, but now it is used as the most dominant version control system in modern software development.

The GPL license of this software ensures its openness, which has enabled a wide variety of individuals worldwide to contribute to its development and usage. Therefore, this software has become a necessity in software development, supporting GitHub and GitLab.

The Linux footprint of this software is evident in its usage and management, which are both effortless processes. The shell scripts developed in this project are a clear example of how Linux command-line tools can be used to automate system processes and work with open-source software.

It is evident that this software, Git, is a clear example of the power of open-source software, which demonstrates its capabilities in building software that benefits the entire global technology community by promoting openness and cooperation among developers worldwide.

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